`DatabaseTokenStore` that stores token state in a new database table in the existing SQL database.

Alternative token storage databases

Although the SQL database storage used in this chapter is adequate for demonstration purposes and low-traffic APIs, a relational database may not be a perfect choice for all deployments. Authentication tokens are validated on every request, so the cost of a database transaction for every lookup can soon add up. On the other hand, tokens are usually extremely simple in structure, so do not need a complicated database schema or sophisticated integrity constraints. At the same time, token state rarely changes after a token has been issued and a fresh token should be generated whenever any security-sensitive attributes change, to avoid session fixation attacks. This means that many uses of tokens are also largely unaffected by consistency worries.

For these reasons, many production implementations of token storage opt for non-relational database backends, such as the Redis in-memory key-value store (https://redis.io), or a NoSQL JSON store that emphasizes speed and availability.

Whichever database backend you choose, you should ensure that it respects consistency in one crucial aspect: token deletion. If a token is deleted due to a suspected security breach, it should not come back to life later due to a glitch in the database. The Jepsen project (https://jepsen.io/analyses) provides detailed analysis and testing of the consistency properties of many databases.

A token is a simple data structure that should be independent of dependencies on other functionality in your API. Each token has a token ID and a set of attributes associated with it, including the username of the authenticated user and the expiry time of the token. A single table is enough to store this structure, as shown in listing 5.2. The token ID, username, and expiry are represented as individual columns so that they can be indexed and searched, but any remaining attributes are stored as a JSON object serialized into a string (varchar) column. If you needed to lookup tokens based on other attributes, you could extract the attributes into a separate